

Homework 2

MATH 518 - Fall 2025

Due on Friday, October 10, 11:59pm on Crowdmark

Total of points = 50 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. Let $I = (x^2 - y^3, y^2 - z^3) \subset k[x, y, z]$ and $V = V(I)$. Define the homomorphism of k -algebras $\alpha: k[x, y, z] \longrightarrow k[t]$ by $\alpha(x) = t^9$, $\alpha(y) = t^6$ and $\alpha(z) = t^4$.
 - (a) (5 points) Show that the kernel of α is I . Deduce that V is irreducible.
 - (b) (5 points) The previous map induces a homomorphism $\bar{\alpha}: k[V] \longrightarrow k[t]$ of k -algebras. What is the morphism $\varphi: k \longrightarrow V$ of affine varieties that corresponds to $\bar{\alpha}$ *i.e.* such that $\varphi^* = \bar{\alpha}$? Show that φ is one-to-one and onto, but not an isomorphism.
2. (10 points) For each of the following functions, determine at which points of the affine variety V they are regular.
 - (a) (5 points) The function $f = \frac{1-\bar{y}}{\bar{x}}$ on $V = V(x^2 + y^2 - 1) \subset k^2$
 - (b) (5 points) The function $f = \frac{\bar{x}}{\bar{y}}$ on $V = V(xw - yz) \subset k^4$.
3. (a) (5 points) Find the singular points of the following curves
 1. $x^4 + y^4 - x^2y^2$
 2. $x^3 + y^3 - 3x^2 - 3y^2 + 3xy + 1$

Hint: simplify the polynomial

 - (b) (5 points) Show that an irreducible plane curve only has finitely many singular points.
4. (10 points) Let $f: k^3 \longrightarrow k^3$ be the regular map $f(x, y, z) = (x, xy, xyz)$. Find the image in k^3 . Is it Zariski closed? open? dense?

Hint: the intersection of two closed sets is closed. If a polynomial $g \in k[x]$ has infinitely many roots then $g = 0$.

5. (10 points) Let $\varphi: V \rightarrow W$ be a morphism of varieties, and $\varphi^*: k[W] \rightarrow k[V]$ the induced k -algebra morphism. Let $p \in V$ and $q \in W$ two points such that $\varphi(p) = q$. Show that φ^* extends uniquely to a ring morphism

$$\varphi^*: \mathcal{O}_q(W) \rightarrow \mathcal{O}_p(V),$$

and that $\varphi^*(\mathfrak{m}_q(W)) \subset \mathfrak{m}_p(V)$.

Remark: In general, φ^ does not extend to morphism $k(W) \rightarrow k(V)$.*